

CO3-AT1-Assignment

View Only

[← Back](#)**QUESTIONS****Q1**Banking Transaction System –
Exception Handling

10 marks

**Q2**Hospital Patient Registration –
User-Defined Exception

10 marks

**Q3**E-Commerce Order Processing
– Built-in Exceptions

10 marks

**Q4**Online Food Delivery – Multiple
Threads and Execution Order

10 marks

**Q5**Manufacturing Plant –
Producer-Consumer Problem**Q3 E-Commerce Order Processing – Built-in Exceptions**

10 marks

Develop a Java program for an online shopping application. Store product names and prices in arrays and accept a product index and quantity from the customer.

The program must:

- Calculate total cost as quantity × price.
- Handle invalid product indices.
- Handle invalid numeric input.
- Reject zero or negative quantities.
- Use appropriate built-in exception classes.
- Continue execution after handling an invalid transaction.

Sample Test Cases:

Input / Scenario Expected Result Purpose

1, 2 Valid product; total cost calculated Normal case

5, 2 `ArrayIndexOutOfBoundsException` handled Boundary / exception case

2, -1 Invalid quantity Invalid input

abc, 2 `InputMismatchException` handled Invalid input

10 marks

5/5 answered

Your Code

```
1 import java.util.*;
2
3 public class Main{
4
5
6 public static void main(String[] args) {
7
8 Scanner sc = new Scanner(System.in);
9     String[] products = {"laptop", "Phone", "Headphon
10     double[] price = {50000, 20000, 3000};
11
12     try{
13         int index = sc.nextInt();
14         int quantity = sc.nextInt();
15
16         if(quantity <= 0){
17             System.out.println("Invalid quantity ")
18
19         } else{
20             double total = price[index]*quantity;
21             System.out.println("Total cost :" +tota
22
23         }
24     }catch(ArrayIndexOutOfBoundsException e) {
25         System.out.println("ArrayIndexOutOfBoundsEx
26     }catch(InputMismatchException e) {
27         System.out.println("InputMismatchException"
28     }
29     System.out.println("Transaction processed");
30 }
```

Input

1 2

Output

```
Total cost :40000.0
Transaction processed
```

31 }

✓ Submitted

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